Stuck on boot logo or automatic repair loop

Article KB0010031 | Owner: Desktop Support | Subcategory: Laptop won't start | Applies to: Dell Latitude 5440/7440, Lenovo ThinkPad T14/X1 Carbon, HP EliteBook 840, Windows 11 23H2 and 24H2

Summary

This article helps users and service-desk agents troubleshoot a Windows laptop that appears stuck on the manufacturer's boot logo, shows a black screen, or enters an endless Automatic Repair loop. It provides a quick self-help hard-reset and startup-repair process, a detailed agent workflow, verification steps, and clear escalation guidance.

Symptoms

- Laptop powers on but remains frozen on the OEM logo (Dell, Lenovo, or HP splash screen)
- Black screen with a blinking cursor after the logo
- Repeated "Preparing Automatic Repair" / "Diagnosing your PC" screens that never reach the login prompt
- Error messages such as "Startup Repair couldn't repair your PC" or "Your PC did not start correctly"

Common causes

- Corrupted or incomplete Windows Update damaging the boot loader
- Faulty early-boot device driver (storage or graphics)
- File system corruption or bad sectors on the SSD/HDD
- Misconfigured BIOS/UEFI settings (Secure Boot, boot order)
- Hardware failure - motherboard, RAM, or SSD

Initial Checks

Before attempting any software fixes, verify that the laptop is receiving power and that the issue is not simply a drained battery or a loose connection. Look for any LED indicators, listen for fan or hard-drive activity, and try connecting to an external monitor to rule out a display-only problem. If the device shows no signs of life, proceed to the hard-reset steps described in the self-help procedure.

Self-Help Procedure: Hard Reset and Startup Repair

Follow these steps to drain residual power, clear temporary boot state, and let Windows attempt its built-in repair. Perform each step in order; if the laptop boots normally at any point, you can stop and resume normal work.

1. Shut down the laptop completely. If it is unresponsive, press and hold the power button for 15 seconds to force a power-off.
2. Disconnect the AC adapter and remove any peripheral devices (USB drives, docks, external monitors).
3. If the model has a removable battery, take it out; otherwise skip this step.
4. With the battery and charger disconnected, hold the power button for 20-30 seconds to discharge residual capacitors.
5. Re-insert the battery (if removed) and reconnect the AC adapter.
6. Press the power button to start the laptop. Observe whether it proceeds past the logo.
7. If the laptop still stops at the logo or shows the Automatic Repair screen, let Windows run Startup Repair automatically (it will appear after a few attempts).
8. When the "Choose an option" screen appears, select "Troubleshoot" -> "Advanced options" -> "Startup Repair" and follow the prompts.
9. After Startup Repair finishes, click "Restart" and check if the system boots to the login screen.
10. If the problem persists, note any specific error codes displayed and contact the IT Service Desk for further assistance.

Service-Desk Agent Procedure

Agents should follow a structured workflow to diagnose and resolve boot-logo or Automatic Repair loop issues. Begin with remote checks, then proceed to hands-on troubleshooting if the user is on-site or has shipped the device. Document each step and any error messages in the ticket.

1. Verify the user's identity and confirm the device model and Windows build via Intune or the asset database.

2. Ask the user to perform the hard-reset steps (as above) and report whether the logo changes or the repair loop persists.
3. If the user cannot perform the steps, arrange a remote session using Teams or request the device be brought to the nearest desktop-support desk.
4. On-site, connect the laptop to AC power and observe LED/fan activity. Attempt to boot to BIOS by pressing the appropriate key (F2 for Dell/Lenovo, F10 for HP) during power-on.
5. In BIOS, check that the internal drive is detected, Secure Boot is enabled (or temporarily disabled for testing), and the boot order lists the Windows Boot Manager first.
6. Exit BIOS and attempt a normal boot. If the logo freeze persists, run the built-in diagnostics (Dell ePSA, Lenovo Vantage, HP PC Hardware Diagnostics) from the BIOS menu.
7. Should diagnostics pass, boot to a Windows Recovery Environment (WinRE) USB drive created from the Microsoft Media Creation Tool.
8. From WinRE, choose "Troubleshoot" -> "Command Prompt" and run `chkdsk C: /f /r` to repair file-system errors, then `sfc /scannow` to check system-file integrity.
9. If chkdsk reports unrecoverable errors or the SSD fails diagnostics, suspect hardware failure; initiate a warranty RMA and issue a loaner laptop per Desktop Support policy.
10. If the file-system repairs succeed, reboot normally and confirm the user can log in. Advise the user to monitor for recurrence and to install pending Windows updates only after verifying stability.

Verification and Closure

After completing either the self-help or agent procedure, confirm that the laptop is functional before closing the ticket. This includes checking that the operating system loads, that the user can access corporate resources (Outlook, Teams, OneDrive), and that BitLocker/FileVault encryption remains intact with the recovery key retrievable from Intune.

1. Log in with the user's corporate credentials and verify that the desktop appears without delay.
2. Open Company Portal and confirm that the device is compliant and that encryption status shows "Protected".
3. Have the user send a test email and open a Teams chat to ensure network connectivity.
4. If any encryption warnings appear, retrieve the BitLocker recovery key from Intune and provide it to the user.
5. Document the final outcome, any parts replaced, and close the ticket with the appropriate resolution code (e.g., "Fixed - Hard reset", "Fixed - Startup repair", "Replaced - Motherboard warranty").

When to escalate

Escalate to Desktop Support L2/Hardware Warranty when: (a) BIOS/UEFI diagnostics report a hardware fault (e.g., SSD read errors, RAM failure), (b) repeated startup repair attempts fail despite a clean WinRE environment, or (c) the device is out of warranty and requires a depot repair. Include in the escalation the BIOS version, SSD model, any error codes from diagnostics, the steps already performed, and the user's contact information for loaner coordination.

Keywords: boot logo stuck, automatic repair loop, laptop won't start, black screen after logo, hard reset, startup repair, UEFI diagnostics, BitLocker recovery, warranty RMA